water gets increased, the wax gets melted and expands the bottom of steel pin in the opposite direction and pushes the seal. By compressing the seal against spring force in the container, the container valve opens. When container gets cooled the wax shrinks and reduces the pressure acting on the steel pin and the container comes to its original position. Now the container valve gets closed.

7.6 WATER PUMP

The water pump is placed in between in front of the engine's cylinder block and the radiator. The Pump impeller consists of a shaft, bearings and water seal which are placed inside the housing. Impeller is fixed to the shaft. The impeller has flat or bended vanes with round shape. Water seal is placed to arrest the leakage with the help of bearings.

Working:

When the impeller rotates, the water in between the blades are expelled out due to centrifugal force. The expelled water moves to the engine water jackets through the pipe with high pressure. The power to the impeller is given from the V-pulley arrangement. The water pump increases the circulation of cooling water. Fig 7.6 shows the water pump.

7.7 ENGINE FAN

When the vehicle runs, natural air is enough to cool the water. However during heavy load and during stationary conditions, air from the atmosphere coming by natural means to the radiator is not sufficient to cool the water. Hence fan is required to supply more air to radiator and to engine. Fan will be coupled with water pump pulley. The fan sucks more air and passes to the radiator core and then cools the water. There are many types of cooling fans used, which include,



Figure 7.7 Engine fan

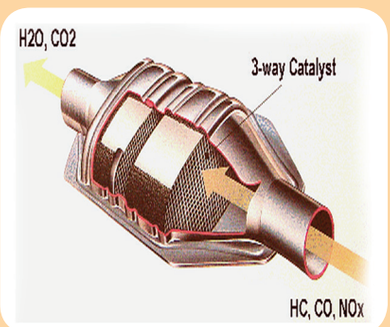


What is catalytic converter?

Catalytic converter was invented by Eugene Houdry in 1930.

Inside the converter, the gases flow through a dense honeycomb structure made from a ceramic and coated with the catalysts.

It reduces harmful NOx and CO emissions into harmless one.



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